



Bajaj TORQ campus challenge 2026

Problem statement

Universal flexible fixture for motorcycle engines (125 - 400 cc)

Team **GMT**

Members :

Anurag Mishra

Aditya Gaddam

Tejas D Rao

Institution :

Visvesvaraya National Institute of Technology Nagpur (VNIT)



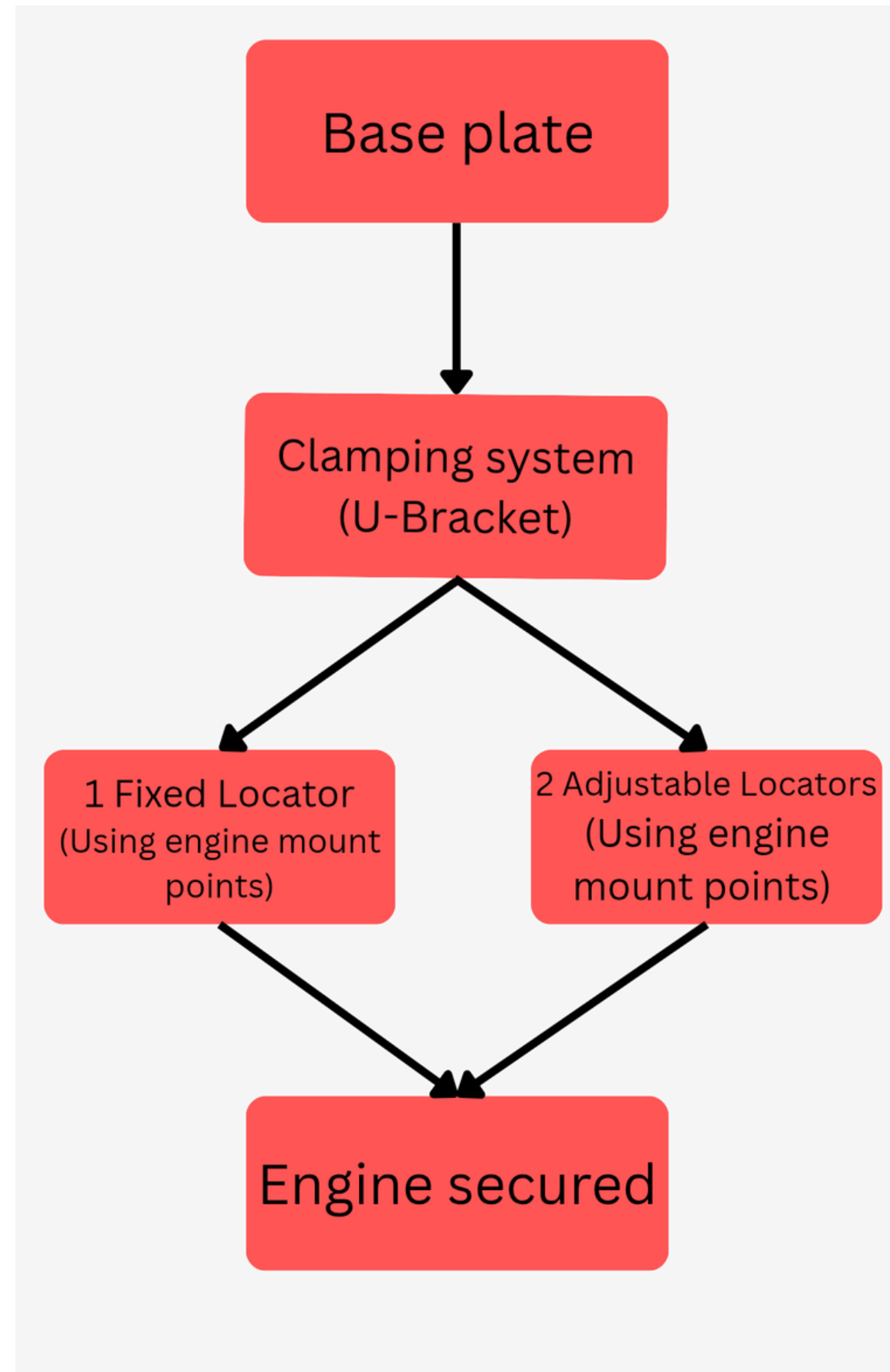
Concept Definition

The fixture developed is considered to be made from mild steel, uses a rigid circular plate, U bracket, one fixed locator and 2 multi point locators, these are fixed to the original engine mounting points that are used to mount the engine on the vehicle, as each engine in the Bajaj lineup of vehicles have atleast 3 engine mounting points this system can be used on any engine varying from 125cc to 400cc.

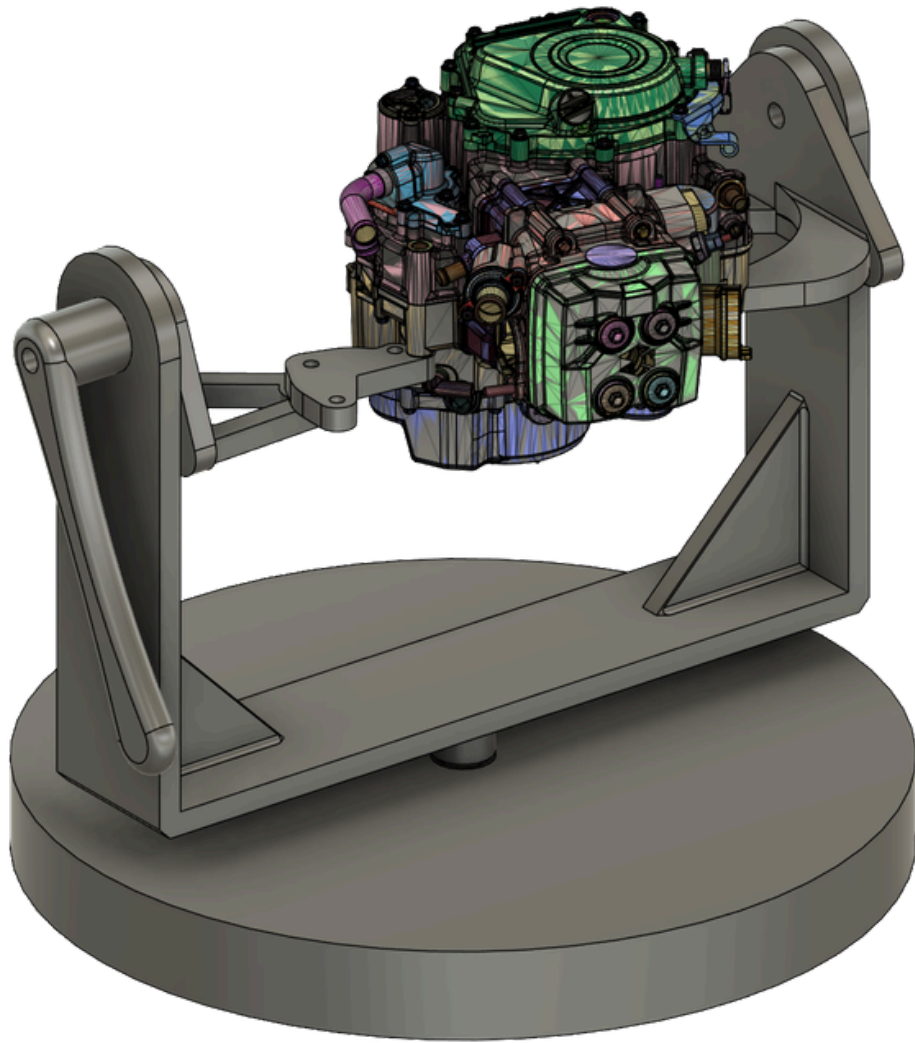
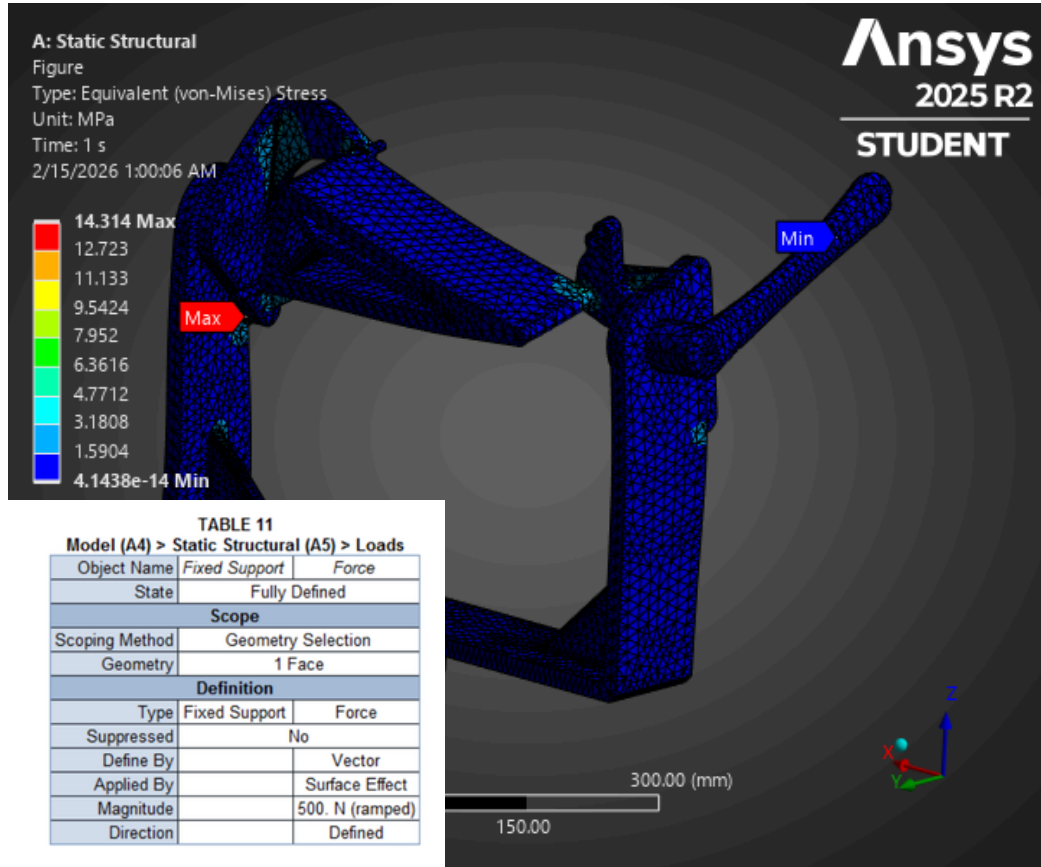
The 3 engine mounting points often form a triangle, the upper most (fixed locator) point is primarily fastened to the fixture and then the other two locators (adjustable locators) are set to the type of engine being worked on and the other two point are also fastened securing the engine from having any kind of motion. The locators are connected to the U bracket through rotating joints which allow the rotation of the engine, enabling the work to be carried out on both the sides of the engine, this can be done by using a lever. The U bracket can also rotate with respect to the rigid plate below it.

The adaptability is increased without harming the accuracy as we have first aligned the top most point of the three engine mounting points. Then the points of all engines that are closer to the first point are aligned along a circular arc which can be connected through a pivot link and the third point is also connected using another pivot link which has been customized to work with all the engines.

All the plates that are involved in the design of this fixture have a width of 20 mm.



Force calculations



Key assumptions

The engine would have contact with the locators at the original engine mounting points, where the centres of both points are aligned

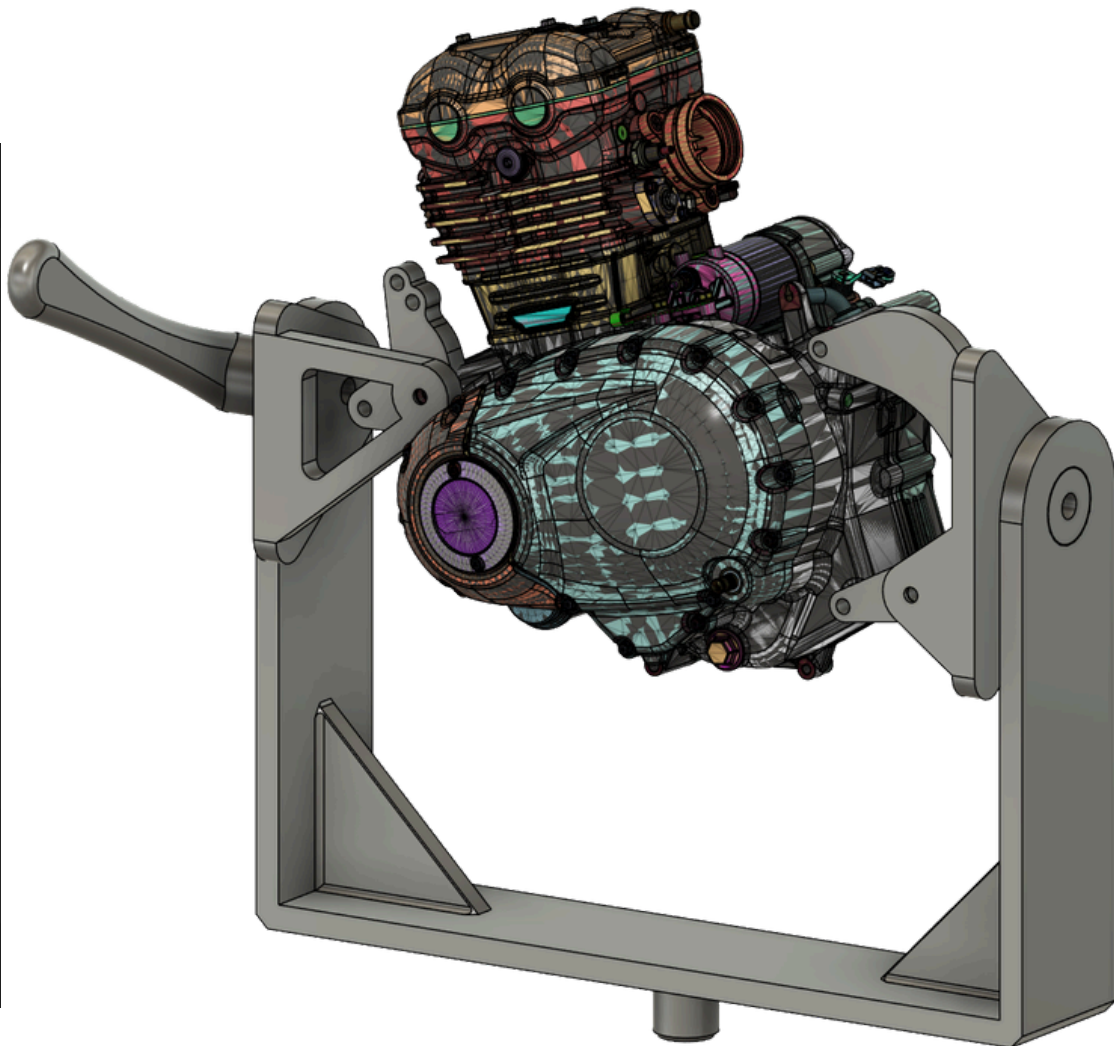
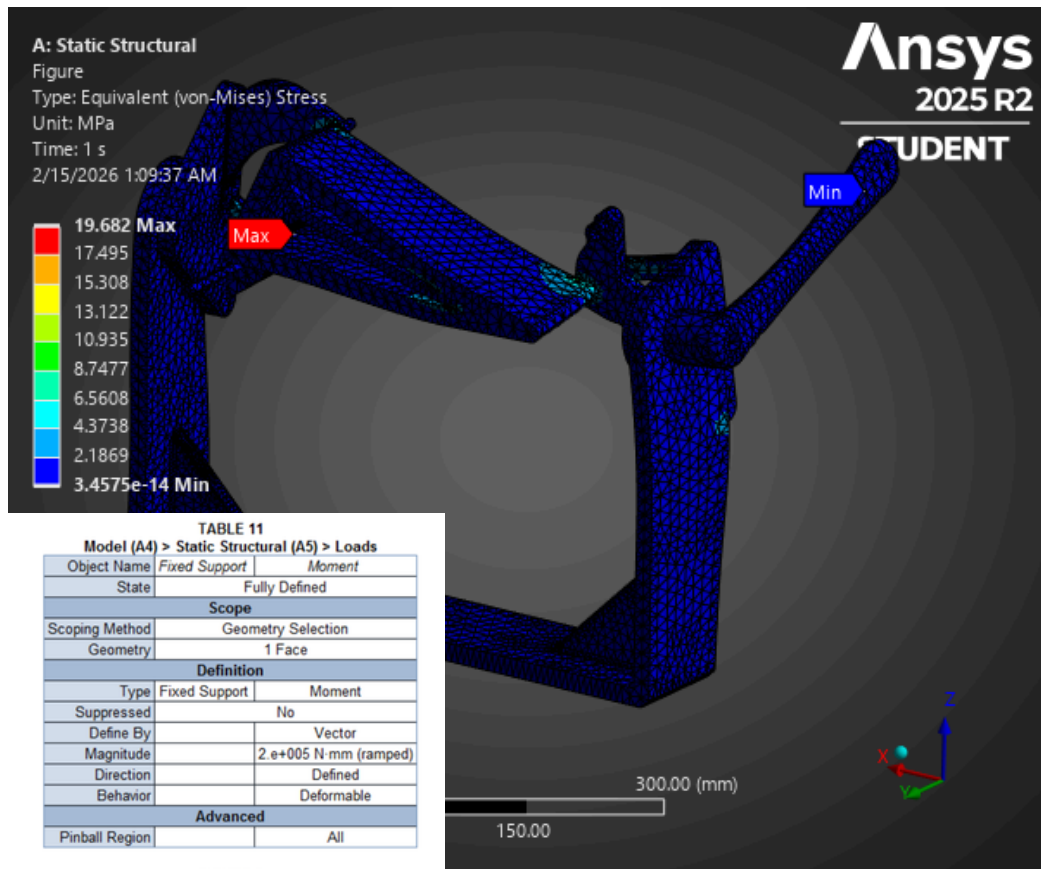
The engine used to compute the calculations resembles a triangular prism and the load is assumed to be disturbed along the face that is away from the locators

The moment is assumed to be applied at the centre of this prism

Stress calculations

The stress values observed under given load conditions are far lower than the yield strength of the material assumed to be used in all possible orientations

Moment calculations



Industrial application & Manufacturability

The fixture designed can be used to work on multiple engines on the same assembly line. The U bracket can be used as an universal part for single cylinder engines and only the custom designed locator needs to be changed if work is to be carried out on a different engine.